Illustrative Target Biology Review

Project AURORA | TAA-17 | Version 1.2

Synthetic demonstration data. This document exists only to show how heterogeneous evidence can be linked to a decision. It is not a real programme or a scientific assertion about a real target.

Decision relevance

The candidate-nomination question is whether TAA-17 biology and a practical patient-selection strategy support advancement of A17-03, while retaining uncertainty about expression heterogeneity, target shedding and normal-tissue binding.

Table 1. Illustrative expression summary

Sample group	n	High / moderate	Low / negative	Interpretation
Ovarian tumour tissue	50	34 (68%)	16 (32%)	Supports a biomarker-defined population; heterogeneity remains material.
Normal epithelial panel	22 tissues	0 high	Low-level signal in corneal epithelium	Requires exposure-margin and functional assessment.
Activated stromal panel	8 donors	3 low/moderate	5 negative	Potential on-target / off-tumour risk in activated fibroblasts.

Figure 1. Internalisation and target engagement

Model	Expression	Internalisation t½	Interpretation
AUR-01	High	36 min	Rapid internalisation
AUR-04	High	45 min	Rapid internalisation
AUR-07	Moderate	44 min	Intermediate
AUR-11	High	43 min	Rapid but incomplete
AUR-15	Low	>120 min	Insufficient for robust activity

Median internalisation half-time across the four high/moderate models was 42 minutes. The low-expression model did not support efficient uptake.

Uncertainty and biomarker readiness

• **Expression heterogeneity:** the proposed programme case applies to a biomarker-defined segment rather than an unselected population.

• **Target shedding:** soluble ectodomain was detectable in 7 of 20 exploratory plasma samples; the exposure consequence is unknown.

• **Biomarker assay:** the prototype IHC classifier achieved an illustrative cross-validated AUC of 0.86, but cut-off and inter-laboratory reproducibility are not locked.

• **Normal-tissue signal:** low-level binding in corneal epithelium and activated fibroblasts requires integration with exposure and functional safety evidence.

Decision-facing conclusion

The biology supports continued development only as a biomarker-defined programme. Candidate nomination should preserve target heterogeneity, shedding and normal-tissue binding as explicit uncertainties, with named studies and reconsideration triggers rather than a single aggregate confidence score.

Source lineage

Evidence object	Underlying source	Owner	Review state
E-001 / E-002	Synthetic IHC tissue panel TB-017	Discovery Biology	Accepted
E-003	Synthetic internalisation study INT-042	Cell Biology	Accepted
E-004	Synthetic soluble-target assay SHED-011	Translational	Awaiting review
E-005	Synthetic biomarker-validation notebook BMK-009	Pathology	Accepted